Sign Language Recognition

Submitted by: Adrutha Leelalakshmi

Reg.No.: 25BCE10630

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School: SCOPE

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INTRODUCTION:

Sign language

Also known as **signed languages**, are languages that use the visual-manual modality to convey meaning, instead of spoken words. Sign languages are expressed through manual articulation in combination with non-manual markers. Sign languages are full-fledged natural languages with their own grammar and lexicon. Sign languages are not universal and are usually not mutually intelligible, although there are similarities among different sign languages.

Communication

Communication is a fundamental part of human interaction, yet millions of people face barriers in everyday communication due to hearing and speech impairments. Sign languages serve as the primary mode of communication for many deaf and hard-of-hearing individuals. However, a large portion of the general population does not understand sign language, which can make daily interactions such as asking for help, communicating in public places, or accessing services difficult. This communication gap often leads to social isolation and reduced accessibility for members of the deaf community.

AI Model

The aim of this project is to develop an AI-based bot capable of translating sign language gestures into text. The system uses image processing and machine learning algorithms to recognize hand signs captured through a camera. Once a gesture is identified, the AI model interprets it and converts it into a corresponding alphabet, or word. This allows users who do not understand sign language to communicate more easily with individuals who rely on it.

Overall, this project highlights the practical application of artificial intelligence in solving real-world problems and showcases how machine learning and computer vision can work together to build assistive technologies.

PROBLEM STATEMENT

- Most people do not understand sign language, creating a communication barrier with deaf or hard-of-hearing individuals.
- Sign language interpreters are not always available in everyday situations.
- This lack of communication tools can make daily interactions difficult and limit accessibility.
- There is a need for an AI-based system that can recognize sign language gestures and convert them into text.
- Such a system can help bridge the communication gap and improve inclusivity.

PROBLEM SOLVING APPROACH

The approach used in this project involves building a machine learning model capable of recognizing sign language gestures through images captured from a camera.

The main steps in the approach include:

- **Data Preparation**

A dataset containing images of different sign language gestures was collected/ prepared and organized into labelled categories.

- **Image processing**

The collected image set was processed i.e. cropped and resized.

- **Model Training**

An ML based model was trained, so it could recognize patterns in hand gestures.

- **Prediction System**

The trained model was integrated with a system that captures images through a camera and predicts the corresponding sign language gesture.

- **Output Translation**

The predicted gesture is converted into corresponding alphabet so that non-sign language users can understand.

KEY DECISIONS MADE

During development several decisions we made, such as

- Using ML and CV to recognize gestures from images.
- Limiting initial model to smaller dataset (images of gestures) for reduced complexity and better efficiency.
- Setting a confidence threshold so predictions are only displayed when the model is confident enough.
- Using Camera-based input system so that the model can detect gestures in real time.

CHALLENGES FACED

Some challenges encountered while development of this project was:

- **Dataset limitations:** Collecting enough labelled images for each gesture was difficult.
- **Model accuracy:** Gesture recognition was be affected by lighting conditions, hand position, and camera quality.
- **Prediction errors:** Sometimes the model may misclassify gestures if they look similar.

WHAT I LEARNED

- How machine learning models are trained using labelled datasets.
- The importance of pre-processing data before training a model.
- How CV techniques allow machines to interpret images.
- The role of debugging and testing in building reliable systems.
- How AI can be applied to solve real-world accessibility problems.
- In-depth knowledge of sign-language and how to use basic sign language

CONCLUSION

The AI-based sign language translation bot demonstrates how ML and CV can be used to address communication barriers. By recognizing hand gestures and converting them into text, the system aims to make communication between sign language users and non-sign language users easier.

Although the current implementation may support a limited number of gestures, it provides a foundation for future improvements. With larger datasets and more advanced models, such systems could eventually support full sign language translation in real time.

Overall, this project highlights the potential of AI to create assistive technologies that promote inclusivity and accessibility in society.